

81. If the integral $525 \int_0^{\frac{\pi}{2}} \sin 2x \cos^{\frac{11}{2}} x \left(1 + \cos^{\frac{5}{2}} x\right)^{\frac{1}{2}} dx$ is equal to $(n\sqrt{2} - 64)$, then n is equal to _____.

यदि समाकलन $525 \int_0^{\frac{\pi}{2}} \sin 2x \cos^{\frac{11}{2}} x \left(1 + \cos^{\frac{5}{2}} x\right)^{\frac{1}{2}} dx = (n\sqrt{2} - 64)$ है, तो n बराबर है _____.

Ans. 176

Sol. $I = 525 \int \sin 2x \cos^{\frac{11}{2}} x \left(1 + \cos^{\frac{5}{2}} x\right)^{\frac{1}{2}} dx$

put $\sqrt{\cos x} = t \Rightarrow \cos x = t^2$

$-\sin x dx = 2t dt$

$\sin x dx = -2t dt$

$$I = 525 \int_0^1 2(2t) t^2 t^{11} \sqrt{1+t^5} dt$$

$$= 525 \times 4 \int_0^1 \sqrt{1+t^5} \cdot t^{14} dt$$

$$= 2100 \int_0^1 \sqrt{1+t^5} \cdot t^{14} dt$$

again put $\sqrt{1+t^5} = u \Rightarrow 1+t^5 = u^2$
 $5t^4 dt = 2u du$

$$I = 2100 \int_1^{\sqrt{2}} u \cdot \frac{2u}{5} (u^2 - 1)^2 du$$

$$= 420 \times 2 \left(\int_1^{\sqrt{2}} u^2 (u^4 - 2u^2 + 1) du \right)$$

$$= 840 \int_1^{\sqrt{2}} (u^6 - 2u^4 + u^2) du$$

$$= 840 \left[\frac{u^7}{7} - \frac{2u^5}{5} + \frac{u^3}{3} \right]_1^{\sqrt{2}}$$

$$= 840 \left[\frac{8\sqrt{2}}{7} - \frac{8\sqrt{2}}{5} + \frac{2\sqrt{2}}{3} - \frac{1}{7} + \frac{2}{5} - \frac{1}{3} \right]$$

$$= 176\sqrt{2} - 64$$

Compare with $n\sqrt{2} - 64$

$$n = 176$$

82. $\left| \frac{120}{\pi^3} \int_0^\pi \frac{x^2 \sin x \cos x}{\sin^4 x + \cos^4 x} dx \right|$ is equal to _____.

$\left| \frac{120}{\pi^3} \int_0^\pi \frac{x^2 \sin x \cos x}{\sin^4 x + \cos^4 x} dx \right|$ बराबर है _____

Ans. 15

Sol. $I = \int_0^\pi \frac{x^2 \sin x \cos x}{\sin^4 x + \cos^4 x} dx$

$$2I = \int_0^\pi \frac{\sin x \cos x}{\sin^4 x + \cos^4 x} (x^2 - (\pi - x)^2) dx$$

$$2I = \int_0^\pi \frac{\pi(2x - \pi) \sin x \cos x}{\sin^4 x + \cos^4 x} dx$$

$$\Rightarrow I = \pi \int_0^{\pi/2} \frac{(2x - \pi) \sin x \cos x}{\sin^4 x + \cos^4 x} dx$$

$$\Rightarrow I = \pi \int_0^{\pi/2} \frac{-2x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$$

$$2I = -\pi^2 \int_0^{\pi/2} \frac{\sin x \cos x}{\sin^4 x + \cos^4 x} dx$$

$$I = \frac{-\pi}{2 \times 2} \int_0^{\pi/2} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$$

$$I = \frac{-\pi^2}{4} \int_0^{\pi/2} \frac{2 \tan x \sec^2 x}{\tan^4 x + 1} dx$$

$$I = \frac{-\pi^2}{4} \tan^{-1} (\tan^2 x) \Big|_0^{\pi/2}$$

$$\Rightarrow I = \frac{-\pi^3}{8}$$

$$\therefore \left| \frac{120}{\pi^3} \right| = 15$$

83. If $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2} \cos x dx}{(1+e^{\sin x})(1+\sin^4 x)} = \alpha\pi + \beta \log_e (3+2\sqrt{2})$, where α, β are integers, then $\alpha^2 + \beta^2$ equals _____.

यदि $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2} \cos x dx}{(1+e^{\sin x})(1+\sin^4 x)} = \alpha\pi + \beta \log_e (3+2\sqrt{2})$ है, जहाँ α, β पूर्णांक हैं, तो $\alpha^2 + \beta^2$ बराबर है _____

Ans. 8

Sol. $I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2} \cos x}{(1+e^{\sin x})(1+\sin^4 x)} dx \quad \dots(1)$

Applyking

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2} \cos x (e^{\sin x})}{(1+e^{\sin x})(1+\sin^4 x)} dx \quad \dots(2)$$

adding (1) & (2)

$$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2} \cos x}{1+\sin^4 x} dx$$

$$I = \int_0^{\frac{\pi}{2}} \frac{8\sqrt{2} \cos x}{1+\sin^4 x} dx$$

$$\sin x = t$$

$$I = \int_0^1 \frac{8\sqrt{2}}{1+t^4} dx$$

$$I = 4\sqrt{2} \int_0^1 \left(\frac{1 + \frac{1}{t^2}}{t^2 + \frac{1}{t^2}} - \frac{1 - \frac{1}{t^2}}{t^2 + \frac{1}{t^2}} \right) dt$$

$$I = 4\sqrt{2} \int_0^1 \frac{\left(1 + \frac{1}{t^2}\right)}{\left(t - \frac{1}{t}\right)^2 + 2} - \frac{\left(1 - \frac{1}{t^2}\right)}{\left(t + \frac{1}{t}\right)^2 - 2} dt$$

$$\text{Let } t - \frac{1}{t} = z \text{ \& } t + \frac{1}{t} = k$$

$$= 4\sqrt{2} \left[\int_{-\infty}^0 \frac{dz}{z^2 + 2} - \int_{\infty}^2 \frac{dk}{k^2 - 2} \right]$$

$$= 4\sqrt{2} \left[\frac{1}{\sqrt{2}} \tan^{-1} \frac{z}{\sqrt{2}} \right]_{-\infty}^0 - \left[\frac{1}{2\sqrt{2}} \ell n \left(\frac{k - \sqrt{2}}{k + \sqrt{2}} \right) \right]_{\infty}^2$$

$$= 4\sqrt{2} \left[\frac{\pi}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} \left[\ell n \frac{2 - \sqrt{2}}{2 + \sqrt{2}} \right] \right]$$

$$= 2\pi + 2\ell n(3 + 2\sqrt{2})$$

$$\alpha = 2$$

$$\beta = 2$$

84. Let $f: (0, \infty) \rightarrow \mathbb{R}$ and $F(x) = \int_0^x t f(t) dt$. If $F(x^2) = x^4 + x^5$, then $\sum_{r=1}^{12} f(r^2)$ is equal to _____.

माना $f: (0, \infty) \rightarrow \mathbb{R}$ तथा $F(x) = \int_0^x t f(t) dt$ हैं। यदि $F(x^2) = x^4 + x^5$ है, तो $\sum_{r=1}^{12} f(r^2)$ बराबर है _____

Ans. 219

Sol. $F(x) = \int_0^x t f(t) dt$ and $F(x)^2 = x^4 + x^5$

$$F'(x) = xf(x) \text{ and } 2xF'(x^2) = 4x^3 + 5x^4$$

$$2F'(x^2) = 4x^2 + 5x^3$$

$$2F'(x) = 4x + 5x^{\frac{3}{2}}$$

$$\Rightarrow 2x + \frac{5}{2}x^{\frac{3}{2}} = xf(x)$$

$$\Rightarrow f(x) = 2 + \frac{5}{2}x^{\frac{1}{2}}$$

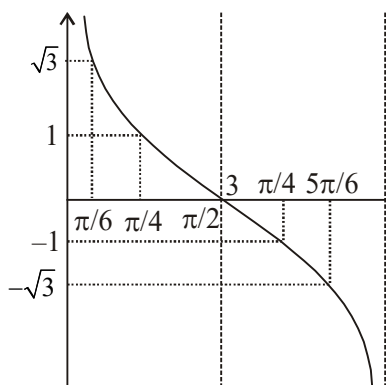
$$\Rightarrow f(x^2) = 2 + \frac{5}{2}x$$

$$\begin{aligned} \sum_{r=1}^{12} f(r^2) &= \sum_{r=1}^{12} \left(2 + \frac{5}{2}r \right) \\ &= 24 + \frac{5}{2} \sum_{r=1}^{12} r \\ &= 24 + \frac{5}{2} \times \frac{12 \times 13}{2} \\ &= 24 + 195 = 219 \end{aligned}$$

85. Let $[t]$ denote the greatest integer $\leq t$. Then $\frac{2}{\pi} \int_{\pi/6}^{5\pi/6} (8[\operatorname{cosec} x] - 5[\cot x])dx$ is equal to _____.

माना $[t]$ महत्तम पूर्णांक $\leq t$ है। तो $\frac{2}{\pi} \int_{\pi/6}^{5\pi/6} (8[\operatorname{cosec} x] - 5[\cot x])dx$ बराबर है _____

Ans. 14



Sol.

$$I = \frac{2}{\pi} \int_{\pi/6}^{5\pi/6} (8[\operatorname{cosec} x] - 5[\cot x]) dx$$

$$\text{For } x \in (\pi/6, 5\pi/6)$$

$$\operatorname{cosec} x \in [1, 2)$$

$$[\operatorname{cosec} x] = 1$$

$$I = \frac{2}{\pi} \left[\int_{\pi/6}^{5\pi/6} 8(1) dx - \int_{\pi/6}^{5\pi/6} 5(1) dx - \int_{\pi/6}^{5\pi/6} 5(0) dx - \int_{\pi/6}^{5\pi/6} 5(-1) dx - \int_{\pi/6}^{5\pi/6} 5(-2) dx \right] = 14$$

86. For $m, n > 0$, let $\alpha(m, n) = \int_0^2 t^m (1+3t)^n dt$. If $11\alpha(10, 6) + 18\alpha(11, 5) = p(14)^6$, then p is equal to _____.

$m, n > 0$ के लिए माना $\alpha(m, n) = \int_0^2 t^m (1+3t)^n dt$ है। यदि $11\alpha(10, 6) + 18\alpha(11, 5) = p(14)^6$ तो p बराबर _____

है।

Ans. 32

Sol. $\alpha(m, n) = \int_0^2 t^m (1+3t)^n dt$

If $11\alpha(10, 6) + 18\alpha(11, 5) = p(14)^6$ then P

$$= 11 \int_0^2 \frac{t^{10}}{11} \frac{(1+3t)^6}{1} + 10 \int_0^2 t^{11} (1+3t)^5 dt$$

$$= 11 \left[(1+3t)^6 \cdot \frac{t^{11}}{11} - \int 6(1+3t)^5 \cdot 3 \frac{t^{11}}{11} \right]_0^2 + 18 \int_0^2 t^{11} (1+3t)^5 dt$$

$$= (t^{11} (1+3t)^6)_0^2$$

$$= 2^{11} (7)^6$$

$$= 2^5 (14)^6$$

$$= 32(14)^6$$

87. Let for $x \in \mathbb{R}$, $S_0(x) = x$, $S_k(x) = C_k x + k \int_0^x S_{k-1}(t) dt$, where $C_0 = 1$, $C_k = 1 - \int_0^1 S_{k-1}(x) dx$, $k = 1, 2, 3, \dots$. Then $S_2(3) + 6C_3$ is equal to _____.

माना $x \in \mathbb{R}$, $S_0(x) = x$, $S_k(x) = C_k x + k \int_0^x S_{k-1}(t) dt$ है, जहाँ $C_0 = 1$, $C_k = 1 - \int_0^1 S_{k-1}(x) dx$, $k = 1, 2, 3, \dots$ हैं। तो

$S_2(3) + 6C_3$ बराबर है _____

Ans. 18

Sol. $C_0 = 1$

$$C_1 = 1 - \int_0^1 S_0(x) dx = 1 - \int_0^1 x dx = 1 - \frac{1}{2} = \frac{1}{2}$$

$$S_1(x) = C_1 x + \int_0^x t dt$$

$$= \frac{1}{2}x + \frac{x^2}{2} = \frac{x + x^2}{2}$$

$$C_2 = 1 - \int_0^1 \left(\frac{x + x^2}{2} \right) dx = 1 - \frac{1}{2} \left(\frac{1}{2} + \frac{1}{3} \right) = 1 - \frac{5}{12} = \frac{7}{12}$$

$$S_2(x) = C_2 x + 2 \int_0^x \left(\frac{t + t^2}{2} \right) dt = \frac{7}{12}x + \left(\frac{x^2}{2} + \frac{x^3}{3} \right) = \frac{7x + 6x^2 + 4x^3}{12}$$

$$S_2(3) = \frac{21 + 54 + 108}{12} = \frac{183}{12} = \frac{61}{4}$$

$$C_3 = 1 - \int_0^1 \left(\frac{7x + 6x^2 + 4x^3}{12} \right) dx$$

$$= 1 - \frac{1}{12} \left[\frac{7}{2} + \frac{6}{3} + \frac{4}{4} \right]$$

$$= 1 - \frac{1}{12} \left(\frac{7}{2} + 3 \right) = \frac{24 - 7 - 6}{24} = \frac{11}{24}$$

$$\text{Now, } S_2(3) + 6C_3 = \frac{61}{4} + \frac{11}{4} = \frac{72}{4} = 18$$

88. Let $f_n = \int_0^{\frac{\pi}{2}} \left(\sum_{k=1}^n \sin^{k-1} x \right) \left(\sum_{k=1}^n (2k-1) \sin^{k-1} x \right) \cos x dx$, $n \in \mathbb{N}$. Then $f_{21} - f_{20}$ is equal to _____.

माना $f_n = \int_0^{\frac{\pi}{2}} \left(\sum_{k=1}^n \sin^{k-1} x \right) \left(\sum_{k=1}^n (2k-1) \sin^{k-1} x \right) \cos x dx$, $n \in \mathbb{N}$ है। तो $f_{21} - f_{20}$ बराबर है _____

Ans. 41

Sol.
$$f_n(x) = \int_0^{\frac{\pi}{2}} \left(1 + \sin x + \sin^2 x + \sin^3 x + \dots + \sin^{n-1} x\right) \left(1 + 3 \sin x + 5 \sin^2 x + \dots + (2n-1) \sin^{n-1} x\right) \cos x dx$$

Multiply & divide by $\sqrt{\sin x}$

$$\int_0^{\frac{\pi}{2}} \left((\sin x)^{\frac{1}{2}} + (\sin x)^{\frac{3}{2}} + (\sin x)^{\frac{5}{2}} + (\sin x)^{\frac{7}{2}} + \dots + (\sin x)^{\frac{2n-1}{2}} \right)$$

$$\left(1 + 3 \sin x + 5 \sin^2 x + \dots + (2n-1) \sin^{n-1} x \right) \frac{\cos x}{\sqrt{\sin x}} dx$$

Put $(\sin x)^{1/2} + (\sin x)^{3/2} + (\sin x)^{5/2} + \dots + (\sin x)^{n-1/2} = t$

$$\frac{1}{2} \frac{\left(1 + 3 \sin x + 5 \sin^2 x + \dots + (2n-1) \sin^{n-1} x \right) \cos x dx}{\sqrt{\sin x}} = dt$$

$$f_n = 2 \int_0^n t dt$$

$$f_n = n^2$$

$$f_{21} - f_{20} = (21)^2 - (20)^2$$

$$= 441 - 400$$

$$= 41$$

Differential Equation

89. Let $f: [1, \infty) \rightarrow [2, \infty)$ be a differentiable function. If $10 \int_1^x f(t) dt = 5xf(x) - x^5 - 9$ for all $x \geq 1$, then the value of $f(3)$ is :

माना $f: [1, \infty) \rightarrow [2, \infty)$ एक अवकलनीय फलन है, यदि $10 \int_1^x f(t) dt = 5xf(x) - x^5 - 9$, $\forall x \geq 1$ तब $f(3)$ का मान होगा:

(A) 18

(B) 22

(C) 26

(D) 32

Ans. D

Sol.
$$10 \int_1^x f(t) dt = 5xf(x) - x^5 - 9$$

differentiate w.r.t. x both sides

$$10f(x) = 5f(x) + 5x f'(x) - 5x^4$$

$$\Rightarrow 5f(x) = 5xf'(x) - 5x^4$$

$$\text{Let } y = f(x) \Rightarrow \frac{dy}{dx} = f'(x)$$

$$5y = 5x \frac{dy}{dx} - 5x^4 \Rightarrow \frac{dy}{dx} - \frac{y}{x} = x^3$$

I.F. of this linear differential equation

$$\Rightarrow \text{I.F.} = e^{\int \frac{-1}{x} dx} = e^{-\ln x} = e^{\ln\left(\frac{1}{x}\right)} = \frac{1}{x}$$

$$\Rightarrow y\left(\frac{1}{x}\right) = \int x^3 \cdot \frac{1}{x} dx = \frac{x^3}{3} + C$$

$$\Rightarrow y = \frac{x^4}{3} + Cx, \text{ at } x = 1$$

$$10 \int_1^1 f(t) dt = 5f(1) - 1 - 9 \Rightarrow f(1) = 2$$

$$\Rightarrow f(1) = \frac{1}{3} + C = 2 \Rightarrow C = \frac{5}{3}$$

$$\Rightarrow f(3) = \frac{3^4}{3} + \frac{5}{3} \cdot 3 = 3^3 + 5 = 32$$

90. Let g be a differentiable function such that $\int_0^x g(t) dt = x - \int_0^x tg(t) dt, x \geq 0$ and let $y = y(x)$ satisfy the differential

equation $\frac{dy}{dx} - y \tan x = 2(x+1) \sec x g(x), x \in \left[0, \frac{\pi}{2}\right]$. If $y(0) = 0$, then $y\left(\frac{\pi}{3}\right)$ is equal to :

माना एक अवकलनीय फलन g के लिए $\int_0^x g(t) dt = x - \int_0^x tg(t) dt, x \geq 0$ है और माना $y = y(x)$, अवकल समीकरण $\frac{dy}{dx} - y \tan$

$x = 2(x+1) \sec x g(x), x \in \left[0, \frac{\pi}{2}\right]$ को संतुष्ट करता है। यदि $y(0) = 0$ है, तो $y\left(\frac{\pi}{3}\right)$ बराबर है :

(A) $\frac{2\pi}{3\sqrt{3}}$

(B) $\frac{4\pi}{3\sqrt{3}}$

(C) $\frac{4\pi}{3}$

(D) $\frac{2\pi}{3}$

Ans. C

Sol. $g(x) = 1 - x g(x)$

$$g(x) = \frac{1}{1+x}$$

$$\frac{dy}{dx} - y \tan x = 2 \sec x$$

$$\text{I.F.} = e^{-\int \tan x dx} = \cos x$$

$$y \cos(x) = \int 2 dx + c$$

$$y(0) = 0$$

$$y = \frac{2x}{\cos x}$$

$$y\left(\frac{\pi}{3}\right) = \frac{4\pi}{3}$$

91. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + 3(\tan^2 x)y + 3y = \sec^2 x$, $y(0) = \frac{1}{3} + e^3$. Then

$$y\left(\frac{\pi}{4}\right) \text{ is equal to :}$$

माना अवकल समीकरण $\frac{dy}{dx} + 3(\tan^2 x)y + 3y = \sec^2 x$, $y(0) = \frac{1}{3} + e^3$ का हल $y = y(x)$ है। तो $y\left(\frac{\pi}{4}\right)$ बराबर है :

(A) $\frac{2}{3} + e^3$ (B) $\frac{4}{3}$ (C) $\frac{4}{3} + e^3$ (D) $\frac{2}{3}$

Ans. B

Sol. $\frac{dy}{dx} + (3 \sec^2 x)y = \sec^2 x$

$$\text{IF} = e^{3 \int \sec^2 x dx} = e^{3 \tan x}$$

$$y \cdot e^{3 \tan x} = \int e^{3 \tan x} \cdot \sec^2 x dx + c$$

$$Y e^{3 \tan x} = \frac{e^{3 \tan x}}{3} + c$$

$$x = 0 \text{ and } y = e^3 + \frac{1}{3}$$

$$e^3 + \frac{1}{3} = \frac{1}{3} + C$$

$$C = e^3$$

$$ye^{3 \tan x} = \frac{e^{3 \tan x}}{3} + e^3$$

$$x = \frac{\pi}{4}$$

$$ye^3 = \frac{e^3}{3} + e^3$$

$$y = \frac{4}{3}$$

92. Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt$ for all $x \in [0, \infty)$.

Then the area of the region bounded by $y = f(x)$ and the coordinate axes is :

माना एक अवकलनीय फलन $f: [0, \infty) \rightarrow \mathbb{R}$ के लिए $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt$, $x \in [0, \infty)$ है। तो $y = f(x)$ और निर्देशांक अक्षों से परिबद्ध क्षेत्र का क्षेत्रफल है :

- (A) $\sqrt{2}$ (B) 2 (C) $\sqrt{5}$ (D) $\frac{1}{2}$

Ans. D

Sol. $f: [0, \infty) \rightarrow \mathbb{R}$

$$f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt \quad \dots(1) \quad \forall x \in [0, \infty)$$

$$= 1 - 2x + e^x \int_0^x e^{-t} f(t) dt$$

$$f'(x) = -2x + e^x \int_0^x e^{-t} f(t) dt + e^x e^{-x} f(x)$$

$$= -2 + f(x) - 1 + 2x + f(x)$$

$$\frac{dy}{dx} = -3 + 2y + 2x$$

$$\frac{dy}{dx} - 2y = 2x - 3$$

$$P = -2; Q = 2x - 3$$

$$\text{I.F.} = e^{-2x}$$

$$y \cdot e^{-2x} = \int e^{-2x} (2x - 3) dx$$

$$y \cdot e^{-2x} = \frac{e^{-2x}}{-2} (2x - 3) - 2 \cdot \frac{e^{-2x}}{4} + C$$

$$y = \frac{3 - 2x}{2} - \frac{1}{2} + C \cdot e^{2x}$$

$$\text{Now, } f(0) = 1$$

$$1 = \frac{3}{2} - \frac{1}{2} + C$$

$$C = 0$$

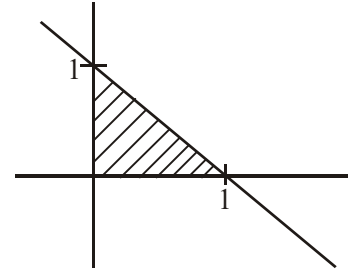
$$y = \frac{3 - 2x - 1}{2}$$

$$= \frac{2 - 2x}{2}$$

$$y = 1 - x$$

$$A = \frac{1}{2} \times 1 \times 1$$

$$= \frac{1}{2}$$



93. If a curve $y = y(x)$ passes through the point $\left(1, \frac{\pi}{2}\right)$ and satisfies the differential equation

$$(7x^4 \cot y - e^x \operatorname{cosec} y) \frac{dx}{dy} = x^5, x \geq 1, \text{ then at } x = 2, \text{ the value of } \cos y \text{ is :}$$

यदि एक वक्र $y = y(x)$ बिन्दु $\left(1, \frac{\pi}{2}\right)$ से होकर जाता है और अवकल समीकरण $(7x^4 \cot y - e^x \operatorname{cosec} y) \frac{dx}{dy} = x^5, x \geq 1$, को

सन्तुष्ट करता है, तो $x = 2$ पर $\cos y$ का मान है :

- (A) $\frac{2e^2 - e}{64}$ (B) $\frac{2e^2 + e}{128}$ (C) $\frac{2e^2 - e}{128}$ (D) $\frac{2e^2 + e}{64}$

Ans. C

Sol. $(7x^4 \cot y - e^x \operatorname{cosec} y) \frac{dx}{dy} = x^5$

$$7x^4 \cos y - e^x = x^5 \sin y \frac{dy}{dx}$$

$$\cos y = t$$

$$\frac{7}{x}t - \frac{e^x}{x^5} = \frac{-dt}{dx}$$

$$\frac{dt}{dx} + \frac{7}{x}t = \frac{e^x}{x^5}$$

$$\text{I.F.} = e^{\int \frac{7}{x} dx} = e^{7 \ln x} = x^7$$

$$tx^7 = \int e^x \cdot x^2 dx + c$$

$$\cos y x^7 = x^2 e^x - 2x e^x + 2e^x + c$$

$$\Rightarrow c = -e$$

$$\text{At } x = 2 \Rightarrow \cos y = \frac{2e^2 - e}{128}$$

94. Let $y = y(x)$ be the solution curve of the differential equation $x(x^2 + e^x) dy + (e^x(x-2)y - x^3) dx = 0$, $x > 0$, passing through the point $(1, 0)$. Then $y(2)$ is equal to :

माना अवकल समीकरण $x(x^2 + e^x) dy + (e^x(x-2)y - x^3) dx = 0$, $x > 0$ का हल वक्र $y = y(x)$, बिन्दु $(1, 0)$ से होकर जाता है। तो $y(2)$ बराबर है :

- (A) $\frac{2}{2-e^2}$ (B) $\frac{4}{4+e^2}$ (C) $\frac{4}{4-e^2}$ (D) $\frac{2}{2+e^2}$

Ans. B

Sol. $x(x^2 + e^x)dy + (e^x(x-2)y - x^3)dx = 0$

$$x(x^2 + e^x) \frac{dy}{dx} + e^x(x-2)y = x^3$$

$$\frac{dy}{dx} + \frac{e^x(x-2)}{x(x^2 + e^x)}y = \frac{x^2}{x^2 + e^x}$$

$$\text{I.F.} = e^{\int \frac{e^x(x-2)}{x(x^2 + e^x)} dx} = e^{\int \frac{e^x \left(\frac{1}{x^2} - \frac{2}{x^2} \right)}{1 + \frac{e^x}{x^2}} dx}$$

$$\text{Let } 1 + \frac{e^x}{x^2} = t \Rightarrow \frac{x^2 e^x - e^x 2x}{x^4} dx = dt$$

$$\Rightarrow \text{I.F.} = e^{\ln \left(1 + \frac{e^x}{x^2} \right)} = 1 + \frac{e^x}{x^2}$$

$$\text{Now } y \left(1 + \frac{e^x}{x^2} \right) = \int \frac{x^2}{x^2 + e^x} \cdot \frac{x^2 + e^x}{x^2} dx + C$$

$$y \left(1 + \frac{e^x}{x^2} \right) = x + C$$

Passing through (1, 0)

$$\Rightarrow C = -1$$

$$y = \frac{x-1}{1 + \frac{e^x}{x^2}}$$

$$y(2) = \frac{1}{1 + \frac{e^2}{4}} = \frac{4}{4 + e^2}$$

95. Let $y = y(x)$ be the solution of the differential equation $(x^2 + 1)y' - 2xy = (x^4 + 2x^2 + 1)\cos x$, $y(0) = 1$. Then

$$\int_{-3}^3 y(x) dx \text{ is :}$$

माना अवकल समीकरण $(x^2 + 1)y' - 2xy = (x^4 + 2x^2 + 1)\cos x$, $y(0) = 1$ का हल $y = y(x)$ है। तो $\int_{-3}^3 y(x) dx$ का मान है :

- (A) 36 (B) 18 (C) 24 (D) 30

Ans. C

$$\text{Sol. } (x^2 + 1) \frac{dy}{dx} - 2xy = (x^2 + 1)^2 \cos x$$

$$\Rightarrow \frac{1}{(x^2 + 1)} \frac{dy}{dx} - \frac{2x}{(x^2 + 1)^2} y = \cos x$$

$$\Rightarrow \frac{d}{dx} \left(y \cdot \frac{1}{1 + x^2} \right) = \cos x$$

$$\Rightarrow \frac{y}{1 + x^2} = \sin x + c$$

$$y(0) = 1$$

$$\Rightarrow 1 = c$$

$$\text{So, } y = (\sin x + 1)(x^2 + 1)$$

$$\begin{aligned}\int_{-3}^3 y(x) dx &= \int_{-3}^3 \sin x \cdot (x^2 + 1) dx + \int_{-3}^3 (x^2 + 1) dx \\ &= 0 + 2 \left(\frac{x^3}{3} + x \right) \Big|_0^3 \\ &= 2 \times 12 \\ &= 24\end{aligned}$$

96. Let $f(x) = x - 1$ and $g(x) = e^x$ for $x \in \mathbb{R}$. If $\frac{dy}{dx} = \left(e^{-2\sqrt{x}} g(f(f(x))) - \frac{y}{\sqrt{x}} \right)$, $y(0) = 0$, then $y(1)$ is :

माना $f(x) = x - 1$ और $g(x) = e^x$, $x \in \mathbb{R}$ हैं। यदि $\frac{dy}{dx} = \left(e^{-2\sqrt{x}} g(f(f(x))) - \frac{y}{\sqrt{x}} \right)$, $y(0) = 0$ हैं, तो $y(1)$ बराबर है :

(A) $\frac{1-e^2}{e^4}$ (B) $\frac{2e-1}{e^3}$ (C) $\frac{e-1}{e^4}$ (D) $\frac{1-e^3}{e^4}$

Ans. C

Sol. $P(f(x)) = (x-1)-1 = x-2$
 $g(ff(x)) = e^{x-2}$

Now $\frac{dy}{dx} = e^{-2\sqrt{x}} \cdot e^{x-2} - \frac{1}{\sqrt{x}} y$

$$\frac{dy}{dx} + \frac{1}{\sqrt{x}} \cdot y = e^{-2\sqrt{x}+x-2} \quad \text{---(1)}$$

$$\text{I.F.} = e^{\int \frac{dx}{\sqrt{x}}} = e^{2\sqrt{x}}$$

Solving of equation: $y \cdot e^{2\sqrt{x}} = \int e^{x-2} dx$

$$y \cdot e^{2\sqrt{x}} = e^{x-2} + C$$

$$y \cdot e^{2\sqrt{x}} = e^{x-2} - e^{-2} \quad (\because y(0) = 0)$$

$$y(1) = \frac{e^{-1} - e^{-2}}{e^2} = \frac{e-1}{e^4}$$

97. Let $x = x(y)$ be the solution of the differential equation $y^2 dx + \left(x - \frac{1}{y} \right) dy = 0$. If $x(1) = 1$, then $x\left(\frac{1}{2}\right)$ is :

माना कि $x = x(y)$, अवकल समीकरण $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$ का हल है। यदि $x(1) = 1$ है, तो $x\left(\frac{1}{2}\right)$ है :

- (A) $3 - e$ (B) $\frac{1}{2} + e$ (C) $\frac{3}{2} + e$ (D) $3 + e$

Ans. A

Sol. $\frac{dx}{dy} = \frac{-x}{y^2} + \frac{1}{y^3}$

$$\frac{dx}{dy} + \frac{x}{y^2} = \frac{1}{y^3}$$

$$\text{I.F.} = e^{\int \frac{1}{y^2} dy} = e^{-\frac{1}{y}}$$

Now, solution of differential equation

$$x \cdot e^{-\frac{1}{y}} = \int e^{-\frac{1}{y}} \cdot \frac{1}{y^3} dy + C$$

Put $-\frac{1}{y} = t$

$$\frac{1}{y^2} dy = dt$$

$$x e^{-\frac{1}{y}} = -\int e^t \cdot t dt + C$$

$$x e^{-\frac{1}{y}} = -[te^t - e^t] + C$$

$$x e^{-\frac{1}{y}} = e^t (1 - t) + C$$

$$x e^{-\frac{1}{y}} = e^{-\frac{1}{y}} \left(1 + \frac{1}{y}\right) + C$$

Now, when $y = 1 \Rightarrow x = 1$

We get, $1 \cdot e^{-1} = e^{-1}(1 + 1) + C$

$$C = -\frac{1}{e}$$

$$\text{Now, } x e^{-\frac{1}{y}} = e^{-\frac{1}{y}} \left(1 + \frac{1}{y}\right) - \frac{1}{e}$$

Put $y = \frac{1}{2}$, we get

$$x \cdot e^{-2} = e^{-2}(1 + 2) - e^{-1}$$

$$x = 3 - e$$

98. Let $f(x)$ be a real differentiable function such that $f(0) = 1$ and $f(x + y) = f(x) f'(y) + f'(x) f(y)$ for all $x, y \in \mathbb{R}$. Then

$\sum_{n=1}^{100} \log_e f(n)$ is equal to :

माना कि $f(x)$ एक वास्तविक अवकलनीय फलन इस प्रकार है कि $f(0) = 1$ है और सभी $x, y \in \mathbb{R}$ के लिए $f(x + y) = f(x) f'(y) + f'(x) f(y)$

$f'(x) f(y)$ है। तब $\sum_{n=1}^{100} \log_e f(n)$ बराबर है :

(A) 2406

(B) 5220

(C) 2525

(D) 2384

Ans. C

Sol. Put $x = y = 0$

$$f(0) = 2f'(0) \text{ as } f(0) = 1$$

$$f'(0) = \frac{1}{2}$$

Now, put $y = 0$ in given equation

$$f(x) = f(x) \cdot f'(0) + f(0) \cdot f'(x)$$

$$f(x) = \frac{1}{2} f(x) + f'(x)$$

$$f'(x) = \frac{1}{2} f(x)$$

$$\frac{dy}{dx} = \frac{y}{2}$$

$$\int \frac{dy}{y} = \int \frac{dx}{2}$$

$$\ln y = \frac{x}{2} + C$$

$$f(0) = 1$$

$$0 = 0 + C$$

$$C = 0$$

$$y = e^{\frac{x}{2}}$$

$$\ln y = \frac{x}{2}$$

$$\text{Now, } \sum_{n=1}^{100} \ln f(n) = \frac{1}{2} + \frac{2}{2} + \frac{3}{2} + \dots + \frac{100}{2} = \frac{1}{2} \left[\frac{100 \times 101}{2} \right] = 2525$$

99. If $x = f(y)$ is the solution of the differential equation $(1+y^2) + (x - 2e^{\tan^{-1}y}) \frac{dy}{dx} = 0$, $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with $f(0) = 1$, then $f\left(\frac{1}{\sqrt{3}}\right)$ is equal to :

यदि $x = f(y)$, $f(0) = 1$ के साथ अवकल समीकरण $(1+y^2) + (x - 2e^{\tan^{-1}y}) \frac{dy}{dx} = 0$, $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ का हल है, तो $f\left(\frac{1}{\sqrt{3}}\right)$ बराबर है :

(A) $e^{\frac{\pi}{6}}$

(B) $e^{\frac{\pi}{4}}$

(C) $e^{\frac{\pi}{12}}$

(D) $e^{\frac{\pi}{3}}$

Ans. A

Sol. $x = f(y)$

$$(1+y^2) + (x - 2e^{\tan^{-1}y}) \frac{dy}{dx} = 0$$

$$\frac{dx}{dy} + \underbrace{\frac{1}{1+y^2}}_P x = \underbrace{\frac{2e^{\tan^{-1}y}}{1+y^2}}_Q$$

$$\text{I.F.} = e^{\int P dy} = e^{\int \frac{1}{1+y^2} dy} = e^{\tan^{-1}y}$$

$$\text{Now } x \times e^{\tan^{-1}y} = \int \frac{2e^{\tan^{-1}y}}{1+y^2} \times e^{\tan^{-1}y} dy + C$$

$$= \int \frac{2(e^{\tan^{-1}y})^2}{1+y^2} dy + C$$

$$xe^{\tan^{-1}y} = \int \frac{2e^{2\tan^{-1}y}}{1+y^2} dy + C$$

put $e^{2\tan^{-1}y} = z$

$$e^{2\tan^{-1}y} \times \frac{2}{1+y^2} dy = dz$$

$$xe^{\tan^{-1}y} = \int 1 dz + C$$

$$xe^{\tan^{-1}y} = z + c$$

$$xe^{\tan^{-1}y} = e^{2\tan^{-1}y} + C$$

$$x = e^{\tan^{-1}y} + \frac{C}{e^{\tan^{-1}y}}$$

given $f(0) = 1$

$y = 0, x = 1$

$$1 = e^0 + \frac{C}{e^0}$$

$$C = 0$$

$$x = e^{\tan^{-1}y}$$

Now put $y = \frac{1}{\sqrt{3}}$

$$x = e^{\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)} = e^{\frac{\pi}{6}}$$

100. Let $x = x(y)$ be the solution of the differential equation $y = \left(x - y \frac{dx}{dy}\right) \sin\left(\frac{x}{y}\right)$, $y > 0$ and $x(1) = \frac{\pi}{2}$. Then $\cos(x(2))$ is equal to :

माना कि $x = x(y)$ अवकल समीकरण $y = \left(x - y \frac{dx}{dy}\right) \sin\left(\frac{x}{y}\right)$, $y > 0$ का हल है और $x(1) = \frac{\pi}{2}$ है। तब $\cos(x(2))$ बराबर है :

- (A) $2(\log_e 2)^2 - 1$ (B) $2(\log_e 2) - 1$ (C) $1 - 2(\log_e 2)$ (D) $1 - 2(\log_e 2)^2$

Ans. A

Sol. $y = \left(x - y \cdot \frac{dx}{dy}\right) \cdot \sin\left(\frac{x}{y}\right)$

$$y dy = (x dy - y dx) \cdot \sin \frac{x}{y}$$

$$\frac{y \, dy}{y^2} = \frac{(x \, dy - y \, dx)}{y^2} \cdot \sin\left(\frac{x}{y}\right)$$

$$\int \frac{dy}{y} = \int \sin\left(\frac{x}{y}\right) \cdot d\left(\frac{x}{y}\right)$$

$$\ln y = -\cos \frac{x}{y} + c$$

$$y = 1, \quad x = \frac{\pi}{2}$$

$$0 = -\cos \frac{\pi}{2} + c$$

$$c = 0$$

$$\therefore \ln y = -\cos \frac{x}{y}$$

$$\text{for } y = 2$$

$$\ln 2 = -\cos \frac{x}{2}$$

$$\cos \frac{x}{2} = -\ln 2$$

$$\cos x = 2 \cos^2 \frac{x}{2} - 1$$

$$= 2 \ln^2 2 - 1$$